

# TITLE TODO

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QUARD UM6P Autonomous Drone Engineering Internship, September 7, 2026



University  
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THE UM6P  
**VANGUARD**  
CENTER

#Empowering Minds.

# Outline

- 1 Introduction
- 2 Main Content
- 3 Conclusion
- 4 Q&A

1 Introduction

2 Main Content

3 Conclusion

4 Q&A

- ▶ Brief overview of the topic
- ▶ Objectives or goals of the presentation
- ▶ Structure of the presentation

1 Introduction

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## Unrolling

# More with bullet points

## Unrolling

- ▶ You could also unroll the slides in a sequence.
- ▶ Just added [`<+>`] option to `\begin{itemize}` in this frame.





## Alerts and repeats

- ▶ The `\alert{}` feature can be particularly useful.
- ▶ Like [this](#).
- ▶ You may as well choose the highlighting to stick.
  - Previous one did not, but this one will.
- ▶ Another common slide type involves columns and/or images.
  - We show them in the next slides.



# More with bullet points

## Alerts and repeats

Sections 12.1 through 12.3 of the [Beamer user manual](#) demonstrate many more features, like alerts and repeats.

- ▶ The `\alert{}` feature can be particularly useful.
- ▶ Like this.
- ▶ You may as well choose the highlighting to stick.
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- ▶ Another common slide type involves columns and/or images.
  - We show them in the [next slides](#).

# Images and columns

Adding images works like in normal  $\text{\LaTeX}$ :

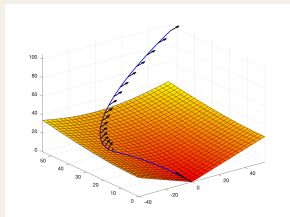
## Code for Adding Images

```
\usepackage{graphicx}  
% ...  
\includegraphics[width=\textwidth]  
{images/uic_quad.pdf}
```



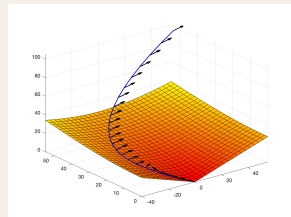
This slide layout was achieved using the `columns` feature. The number and widths of the columns is totally user preference. Another columns example follows on the next slide.

# Another images/columns example



**Figure:** Minimum fuel trajectory

**Left:** Solving for optimal fuel consumption **Right:** Solving for optimal time taken



**Figure:** Minimum time trajectory

Blocks are another great way to partition your slide. They can be combined with columns, as shown here.

## Standard Blocks

For the light theme, their color matches the footline. We will discuss light and dark themes soon.

```
\begin{block}{title}  
content...  
\end{block}
```

## Color Blocks

Similar to the ones on the left, but you pick the color. Text will be white by default, but you may set it with an optional argument.

```
\begin{colorblock}[black]{um6p-dark}{title}  
content...  
\end{colorblock}
```

# Definition & Theorem

## Definition - Group

A *group* is a set  $G$  together with an operation  $\cdot : G \times G \rightarrow G$  such that the following conditions hold:

- ▶ (Associativity) For all  $a, b, c \in G$ , we have  $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ .
- ▶ (Identity) There exists an element  $e \in G$  such that for all  $a \in G$ ,  $a \cdot e = e \cdot a = a$ .
- ▶ (Inverse) For each element  $a \in G$ , there exists an element  $a^{-1} \in G$  such that  $a \cdot a^{-1} = a^{-1} \cdot a = e$ .

## Theorem - Lagrange's Theorem

Let  $G$  be a finite group, and let  $H$  be a subgroup of  $G$ . Then the order (number of elements) of  $H$  divides the order of  $G$ . In other words, if  $|G|$  and  $|H|$  represent the number of elements in  $G$  and  $H$ , respectively, then:

$$|G| = [G : H] \cdot |H|,$$

where  $[G : H]$  is the index of  $H$  in  $G$ , representing the number of distinct left cosets of  $H$  in  $G$ .

# Proposition, Corollary, Example & Remark

## Proposition

Let  $G$  be a group, and let  $e$  be its identity element. Then  $e$  is unique, meaning if  $e'$  is another identity element in  $G$ , we have  $e = e'$ .

## Corollary

In any group  $G$ , the inverse of each element is unique.

## Example

Consider the set  $\mathbb{Z}$  of integers under addition. This forms a group where the identity element is 0, and for any  $a \in \mathbb{Z}$ , the inverse of  $a$  is  $-a$ .

## Remark

The associativity property in a group does not imply commutativity. A group is called *abelian* if  $a \cdot b = b \cdot a$  for all  $a, b \in G$ .

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**Algorithm** Bellman-Kalaba (adapted from [algorithmicx documentation](#)).

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```
Input:  $G, u, l, p$ 
for all  $v \in V(G)$  do
   $l(v) \leftarrow \infty$ 
end for
 $l(u) \leftarrow 0$ 
repeat
  for  $i \leftarrow 1, n$  do
     $min \leftarrow l(v_i)$ 
    for  $j \leftarrow 1, n$  do
      if  $min > \text{EDGE}(v_i, v_j) + l(v_j)$  then
         $min \leftarrow \text{EDGE}(v_i, v_j) + l(v_j)$ 
         $p(i) \leftarrow v_j$ 
      end if
    end for
     $l'(i) \leftarrow min$ 
  end for
   $changed \leftarrow l \neq l'$ 
   $l \leftarrow l'$ 
until  $\neg changed$ 
```

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▷ Example comment

# Tables

| Service                   | Priority | Started | Responsible |
|---------------------------|----------|---------|-------------|
| Internal Messaging        | ♣♣♣      | ✗       |             |
| Notification              | ♣♣♣      | ✓       |             |
| Announcement              | ♣♣♣      | ✗       |             |
| Event Management          | ♣♣♣      | ✗       |             |
| Internal Newsletter       | ♣♣♣      | ✗       |             |
| Feedback Survey           | & ♣      | ✗       |             |
| Podcast & Webinar Archive | ♣        | ✗       |             |

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# Conclusion

- ▶ Summary of key points
- ▶ Call to action or next steps
- ▶ Closing remarks

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Thank you for your attention!  
Any questions?